

```
import pandas as pd
```

```
import numpy as np
```

```
puma = pd.read_csv("/Users/anurag/Downloads/Puma_Sales_Dataset_raw(in).csv")
```

```
# print(puma.info())
```

```
# print(puma.duplicated().sum())
```

```
puma.drop_duplicates(inplace=True)
```

```
# print(puma.duplicated().sum())
```

```
|  
# print(puma["Return_Reason"].unique())
```

```
# print(puma["Return_Reason"].isnull().sum())
```

```
puma["Return_Reason"].fillna("Not Specified", inplace=True)
```

```
# print(puma["Return_Reason"].unique())
```

```
|  
# print(puma["Customer_Rating"].unique())
```

```
# print(puma["Customer_Rating"].isnull().sum())
```

```
puma["Customer_Rating"] = puma["Customer_Rating"].fillna(puma["Customer_Rating"].mode()  
[0])
```

```
# print(np.sort(puma["Customer_Rating"].unique()))
```

```
|  
# print(puma["Currency"].unique())
```

```
|  
# print(puma["Payment_Method"].unique())
```

```
|  
# print(np.sort(puma["Inventory_Available"].unique()))
```

```
|  
# print(puma["Customer_Name"].isnull().sum())
```

```
puma["Customer_Name"].fillna("Unknown", inplace=True)
```

```
# print(puma["Customer_Name"].isnull().sum())
```

```
|  
# print(puma["Gender"].unique())
```

```
# print(puma["Gender"].isnull().sum())
```

```
puma["Gender"].fillna(puma["Gender"].mode()[0], inplace=True)
```

```
# print(puma["Gender"].isnull().sum())
```

```
puma["Gender"] = puma["Gender"].replace({
```

```
    "F":"Female",
```

```
    "M":"Male"
```

```
})
```

```
# print(puma["Gender"].unique())
```

```
|  
# print(np.sort(puma["Age"].unique()))
```

```
puma["Age"] = puma["Age"].fillna(puma["Age"].mode()[0])
```

```
# print(np.sort(puma["Age"].unique()))
```

```
|  
# print(puma["Country"].unique())
```

```
|  
# print(puma["City"].unique())
```

```
puma["City"] = puma["City"].replace({
```

```
    "NewYork" : "New York",
```

```
    "Delhii" : "Delhi"
```

```
})
```

```
# print(puma["City"].unique())
```

```
|  
# print(puma["Store_Name"].unique())
```

```
puma["Store_Name"] = puma["Store_Name"].replace({
```

```
    "Puma-Mumbai" : "Puma Mumbai",
```

```
    "PUMA Mumbai" : "Puma Mumbai"
```

```
} )
```

```
# print(puma["Store_Name"].unique())
```

```
|
```

```
# print(puma["Sales_Channel"].unique())
```

```
|
```

```
# print(puma["Product_Category"].unique())
```

```
|
```

```
# print(puma["Product_Name"].unique())
```

```
|
```

```
# print(np.sort(puma["Quantity"].unique()))
```

```
|
```

```
# print(np.sort(puma["Unit_Price"].unique()))
```

```
|
```

```
# print(np.sort(puma["Discount_%"].unique()))
```

```
|
```

```
puma["Order_Date"] = pd.to_datetime(puma["Order_Date"], format="mixed", errors="coerce")
```

```
# print(puma["Order_Date"].head())
```

```
puma["Order_Date"] = puma["Order_Date"].dt.strftime("%Y-%m-%d")
```

```
# print(puma["Order_Date"].unique())
```

```
|
```

```
puma = puma.drop(columns=["Discount_%"])
```

```
|
```

```
# print(np.sort(puma["Revenue"].unique()))
```

```
|
```

```
puma = puma.drop(columns=["Revenue"])
```

```
puma["Revenue"] = puma["Quantity"] * puma["Unit_Price"]
```

```
# print(np.sort(puma["Revenue"].unique()))
```

```
|
```

```
# print(np.sort(puma["Cost"]))
```

```
# print(np.sort(puma["Profit"]))
```

```
|  
puma = puma.drop(columns="Profit")
```

```
puma["Profit"] = puma["Revenue"] - puma["Cost"]
```

```
# print(np.sort(puma["Profit"]))
```

```
# print(puma.head())
```

```
# print(puma.isnull().sum())
```

```
# print(puma.info())
```

```
puma = puma.astype({
```

```
    "Revenue" : "float64",
```

```
    "Unit_Price" : "float64",
```

```
    "Age" : "int64",
```

```
    "Order_Date" : "datetime64[ns]",
```

```
    "Customer_Rating" : "float64",
```

```
    "Revenue" : "float64"
```

```
})
```

```
puma["Profit"] = puma["Profit"].round(2)
```

```
puma.to_csv("/Users/anurag/Downloads/Puma_Dataset_After_Python.csv", index = False)
```